To the Esteemed Members of the County Steering Group (CSG),

I have had the opportunity to review the "Participatory Rangeland Degradation Assessment Draft Report for Garissa County Government. This is an invaluable foundational document that provides critical insights into the ecological and socioeconomic challenges facing our rangelands. With a few professional refinements, this report can become an even more powerful tool for positive change on the ground.

My review, approached from a geospatial perspective, has highlighted a few key areas where we can enhance the report's accuracy, accessibility, and utility for the communities it aims to serve. These are not criticisms of the effort but rather constructive suggestions to leverage the investments our county has already made in modern technology.

Professional Observations and Recommendations

My primary focus is on ensuring this vital information reaches and is understood by the uneducated pastoralist at the "last mile." For this to happen, the data must be more than just scientifically sound; it must be visually intuitive and relatable to their lived experience.

1. **Ensuring Data Integrity: The Importance of Verified Geospatial Data**

I noticed some minor data discrepancies in the report, such as the elevation of Garissa being cited as **1,138 meters** above sea level. This figure is inconsistent with verified, on-the-ground data and the report's own description of our low-lying plains.

As you know, both the **Garissa County Government** and **Garissa University** have invested heavily in a stable, modern GIS infrastructure powered by ArcGIS Enterprise. This system, which is securely hosted on the cloud by the ICT Authority, provides an authoritative source of truth for all our geospatial data. It includes tools like the **Survey123 data collection toolkit**, which could have been used to feed data directly into our GIS portals for real-time analysis and validation, preventing these inconsistencies before they are even published. This capability is fully functional and can still be utilized to validate the existing data.
For our partners, the GIS portal can be accessed through https://gis.garissa.go.ke/portal/home/. Any partner with relevant data can request access by sending an email to **gis@garissa.go.ke**, and we will be happy to grant it. We invite you all to interact with our data and leverage this resource.

2. **Making Geospatial Data Accessible: Simplified Maps and Storytelling**

The maps in the report, while scientifically accurate in their representation of coordinates and land cover, are too technical for a non-expert audience. A map with an unlabelled axis and abstract symbols does not resonate with a pastoralist who knows their landscape by its rivers, landmarks, and seasonal grazing patterns.

Our GIS infrastructure provides a perfect solution for this through **Story Maps**. This tool allows us to combine interactive maps with photos, videos, and narrative text to tell a compelling story. Instead of a static map, we can create an engaging, visual narrative that shows the community where degradation is happening, why it's a problem, and what they can do about it. The report's own mention of **Participatory Geographic Information Systems (PGIS)** aligns perfectly with this approach. We can develop simplified maps that use clear, relatable symbols and visual language that an uneducated person can easily interpret.

3. **Enhancing Communication: The Power of Imagery and Visuals**

The report's plant species composition table is an excellent start, but a simple list of names, even with their incorrect Somali names, is not enough for an audience that learns visually. To truly convey the scale of the problem and the potential for restoration, the report should incorporate high-pixel imagery.

I strongly recommend updating **Table 23** to include clear photos of all the plant species, highlighting what they look like in both the lush **wet season** and the dry, depleted state. We can also include photos of key indicator species and invasive plants like the Mathenge tree (*Prosopis juliflora*) to make the concepts of land degradation and invasive species management tangible. This visual evidence will provide our community with the ability to "read" their landscape and understand the report's findings without needing to be literate.

For remote sensing and Earth observation, our GIS infrastructure is also linked to **Google Earth Engine**, which provides a powerful platform for analyzing historical satellite data and monitoring changes in real-time. This tool can be used to provide a visual timeline of degradation and recovery.

I also want to highlight the parallel work being done on the **Africa Adaptation Atlas**, which can be accessed at **https://adaptationatlas.cgiar.org/data-explorations/explore-the-datasets**. This system contains open-source data on degradation, drought, and floods that can serve as an excellent baseline for our project. I encourage all members to log in and explore its capabilities.

In conclusion, this report is a commendable effort, and with these few, easily implemented improvements, we can transform it into a truly impactful document. By leveraging our existing GIS infrastructure, we can not only correct minor inconsistencies but also make the findings accessible and actionable for every member of our community, ultimately strengthening our collective efforts towards sustainable rangeland management.

With kind regards,

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